
A TIGHT SECURITY ANALYSIS OF THE FIPS-205 STANDARD (SLH-DSA)

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ABSTRACT

The SPHINCS+ framework, recently standardized by NIST as SLH-DSA (FIPS 205), is a leading stateless hash-based signature scheme for the post-quantum era. Concrete-security evaluation of SPHINCS+ faces a trade-off: tight bounds require a conjectural single function, multi-target decisional second-preimage resistance (SM-DSPR), while fully proven bounds incur substantial looseness. In this paper, we refine the concrete security analysis of SPHINCS+ by eliminating this heuristic reliance. In our approach, we analyze the SM-openPRE and SM-PRE properties instead of relying on SM-DSPR. By utilizing a precise probabilistic simulation technique, we restrict the multi-target tightness degradation exclusively to the maximum number of practically revealed targets rather than the total theoretical targets. When applied to the SLH-DSA parameter sets, our conjecture-free approach bridges the theoretical gap, recovering up to 18 bits of classical security and 9 bits of quantum security compared to the NIST loose evaluation. Our results establish a provable bound for the practical security of SLH-DSA without relying on optimistic multi-target assumptions.

1 Introduction

Digital signature schemes are a foundational primitive for ensuring integrity, authentication, and non-repudiation in global information-security standards (e.g., FIPS, GOST). Digital signatures face an existential threat from large-scale quantum computers running Shor’s algorithm [5], accelerating the global transition to Post-Quantum Cryptography (PQC). To standardize quantum-resistant primitives, NIST initiated a rigorous evaluation process in 2016 [7], culminating in the selection of SPHINCS+—a premier stateless hash-based signature scheme—for standardization (for FIPS 205) [9] in August 2023.

SPHINCS+ fundamentally overcomes the restrictive state-management bottlenecks of its stateful predecessors (e.g., XMSS, LMS). It achieves this by orchestrating a multi-layered Hypertree architecture that seamlessly integrates One-Time Signatures (OTS) and Few-Time Signatures (FTS). This hierarchical structure authenticates a virtually inexhaustible supply of key pairs at the lowest layer, rendering the scheme entirely stateless. Crucially, the post-quantum resilience of SPHINCS+ is formally guaranteed through tight security reductions, rooting its security entirely in the fundamental properties of its underlying cryptographic hash functions.

Although the theoretical security of the SPHINCS+ has been well-established via reductionist proofs in the quantum random oracle model (QROM) [2, 3], its practical deployment demands a more rigorous concrete security evaluation.

First, parameter-dependent security & loose concrete bounds: Existing analyses use conservative adversarial query complexity limits to simplify proofs. While theoretically sound, this yields either loose concrete security estimates or forcing pessimistic parameter selections (n, h, d, k, t) that unnecessarily degrade practical performance. Tightening these bounds, particularly in multi-target settings (where multiple independent targets amplify an adversary’s success probability), is crucial.

Second, reliance on conjectured bounds: Most critically, the concrete security of SPHINCS+ is reduced to the properties of its underlying Tweakable Hash Functions (THFs). However, current evaluations depend on conjectured (heuristic) bounds for specific properties, such as Single-function Multi-target Decisional Second-Preimage Resistance (SM-DSPR) and Single-function Multi-target Preimage Resistance (SM-PRE).

Problem statement: We aim to tighten and clarify concrete security bounds for SPHINCS+-with particular attention to multi-target settings. Therefore, this paper focuses on minimizing reliance on heuristic bounds by establishing mathematically provable evaluation perspectives for these underlying THF properties.

1.1 Our Contributions

In this paper, we bridge the theoretical gaps in the concrete security evaluation of the SPHINCS+ signature scheme. Our main contributions are threefold:

1. Conjecture-Free Security Bound for SM-PRE:

Prior analyses of SPHINCS+ [2, 3] derive the SM-PRE property for the underlying THFs ($\mathbf{Th}$) is bounded via SM-TCR and SM-DSPR. Specifically, the relationship is given by:

$$\text{Succ}_{\mathbf{Th},p}^{\text{SM-PRE}}(\mathcal{A}) \leq 3 \cdot \text{Succ}_{\mathbf{Th},p}^{\text{SM-TCR}}(\mathcal{A}) + \text{Adv}_{\mathbf{Th},p}^{\text{SM-DSPR}}(\mathcal{A})$$

Since the bound for $\text{Adv}_{\mathbf{Th},p}^{\text{SM-DSPR}}(\mathcal{A})$ is conjectural, inherently rendering the overall SM-PRE bound conjectural as well. Kudinov *et al.* [13, 14] recently derived a conjecture-free QROM bound for SM-PRE by reducing it to SM-UD and a newly formalized notion of single-function multi-target one-wayness SM-OW for distinct tweaks. Their provable bound is given by:

$$\text{Adv}_{\mathbf{Th},\mathcal{M}',p}^{\text{SM-PRE}}(\mathcal{A}) \leq \frac{8(q+1)^2}{|\mathcal{H}|} + \frac{12(q+1)}{\sqrt{|\mathcal{M}'|}}.$$

Aligning with the recent trajectory set by Kudinov *et al.* [13, 14] to eliminate conjectural, our first contribution formally resolves this dependence by proposing a provable security evaluation for SM – PRE. However, we take an alternative reduction route: by reducing SM – PRE directly to the well-studied Average-case Search (Avg-Search) problem [1], we derive a non-conjectural, provable bound of

$$\text{Succ}_{\mathbf{Th},p}^{\text{SM-PRE}}(\mathcal{A}) \leq \frac{32(q+1/2)^2}{2^n}.$$

2. Optimized the security bound for SM-openPRE:

We re-analyze the SM-openPRE property for tweakable hash functions in Game 4 (GAME_4) of the SPHINCS+ security proof[2, 3] where the argument originally relied on the T -openPRE property for keyed hash functions. The bound for SM-openPRE is deduced from SM-DSPR and SM-SPR (with p targets) in [6]. If one applies the mathematically proven-yet loose-bound for SM-DSPR from [4] (where $\text{Adv}_{\mathbf{H}}^{T\text{-DSPR}}(\mathcal{A}) = T \cdot \text{Adv}_{\mathbf{H}}^{\text{DSPR}}(\mathcal{B})$), the reduction suffers a massive loss factor of $T = p$ (the total number of targets). To address this, we propose an alternative approach: reducing SM-PRE to SM-openPRE. This technique allows us to trade-off a smaller loss factor $p' \mid p$ instead of the factor p . Consequently, this yields a provable bound with a significantly reduced loss factor and a tighter, fully justified security estimate fo GAME_4 . Table 1 summarizes the comparison between our approach and previous works.

Table 1: Comparision of security bounds for SM-openPRE

Approach	Reduction basis	Loss factor	Remarks
Conjectural [2, 3]	SM-DSPR (conjecture)	≈ 1	Based on a conjecture
Loose bound [4]	T-DSPR (provable)	$T = p$	Proven but large loss p
This work	SM-PRE to SM-openPRE	p' (with $p' \mid p$)	Proven: p' typically $\leq p$

3) Rigorous Parameter Evaluation and Instantiation:

We re-evaluate the standard SPHINCS+ (SLH-DSA) parameters based on our conjecture-free bounds. Consequently, we propose new, optimized parameter sets that achieve NIST’s target security levels without relying on conjectural assumptions.

Organization: The paper is organized as follows. Section 1 gives an introduction and motivation. Section 2 reviews preliminaries and notation used throughout the paper. Sections 3 and 4 present our main technical results: delivering the formal security proofs and delineating our novel perspectives on the security bound evaluation of SPHINCS+. Finally, Section 5 concludes the paper.

2 Preliminaries

Notation: Let $x \xleftarrow{\$} \mathcal{X}$ denote that x is sampled uniformly at random from the set $\mathcal{X}$. We use uppercase letters $\mathcal{A}, \mathcal{B}$ to denote algorithms, and $y \leftarrow \mathcal{A}(x)$ denotes the output of $\mathcal{A}$ on input x . We write $\mathcal{A}^{\mathcal{B}}$ to denote that $\mathcal{A}$ has oracle access to $\mathcal{B}$. We use $\mathcal{A}^{(\mathcal{O})}$ to denote an algorithm $\mathcal{A}$ equipped with quantum access to an oracle $\mathcal{O}$. We say a function is negligible and denote it as $\text{negl}(n)$ if it vanishes faster than any polynomial.

2.1 Tweakable Hash Functions and Security Properties

Definition 1 (Tweakable Hash Function - THF, [2]). *Let $n, m \in \mathbb{N}$. Given a public parameter space $\mathcal{P}$ and a tweak space $\mathcal{T}$, a THF is an efficiently computable function defined as $\text{Th} : \mathcal{P} \times \mathcal{T} \times \{0, 1\}^m \rightarrow \{0, 1\}^n$. It maps an m -bit message M to an n -bit digest MD using a public parameter $P \in \mathcal{P}$ and a tweak $T \in \mathcal{T}$.*

The SPHINCS+ framework [2, 3] specifies two primary THF constructions:

- *THF1 (keyed + masking).* Let $H_1 : \{0, 1\}^{2n} \times \{0, 1\}^m \rightarrow \{0, 1\}^n$ be a keyed hash and a mask generator $H_2 : \{0, 1\}^{2n} \rightarrow \{0, 1\}^m$. For $\mathcal{P} = \mathcal{T} = \{0, 1\}^n$,

$$\text{Th}(P, T, M) = H_1(P \| T, M \oplus H_2(P \| T))$$

- *THF2 (simple, unkeyed).* Let $H : \{0, 1\}^{2n+m} \rightarrow \{0, 1\}^n$ be an unkeyed hash. For $\mathcal{P} = \mathcal{T} = \{0, 1\}^n$, it directly hashes the concatenated inputs:

$$\text{Th}(P, T, M) = H(P \| T \| M)$$

The second construction (THF2) is the one adopted in the standardized instantiation of SPHINCS+, namely SLH-DSA [2].

Security Properties for SPHINCS+: The security of SPHINCS+ relies on several properties of its underlying THFs for distinct tweaks. These include *Single-function Multi-target Target-Collision Resistance* (SM-TCR), *Single-function Multi-target Decisional Second-Preimage Resistance* (SM-DSPR), *Single-Function Multi-Target Preimage Resistance* (SM-PRE), and *Undetectability* (SM-UD). Additionally, [6] introduced *Single-function Open Preimage Resistance* (SM-openPRE). SPHINCS+ also requires *Interleaved Target Subset Resilience* (ITSR) to capture the security of the FORS component.

Table 2 summarizes the known quantum security bounds for these THF properties as utilized in the SPHINCS+. Note that the bounds represent the advantage of a quantum adversary $\mathcal{A}$ making q quantum queries to the function and p classical queries to the challenge oracle. The security parameter n is the digest length of Th , and $X = \sum_{\gamma} (1 - (1 - 1/t)^{\gamma})^k \binom{p}{\gamma} (1 - 1/2^h)^{p-\gamma} \frac{1}{2^{h\gamma}}$.

Table 2: Success probabilities of generic attacks

Property	Success Probability (Quantum)	Security Proof
SM-TCR	$\Theta((q+1)^2/2^n)$	Proven in [2, 3]
SM-DSPR	$\Theta((q+1)^2/2^n)$	Conjectured in [2, 4]
SM-PRE	$\Theta((q+1)^2/2^n)$	Conjectured in [2, 3], Proven in [13, 14]
PRF	$\Theta(12q/\sqrt{2^n})$	Proven in [11]
SM-UD	$\Theta(12q/\sqrt{2^n})$	Proven in [3]
ITSR	$\Theta((q+1)^2 \cdot X)$	Proven in [2]

To formalize the cryptographic strength of the underlying (tweakable) hash functions throughout this paper, we adopt the metric of *insecurity*, denoted by InSec . For a specific security property Prop of a function F involving a parameter p

(which inherently denotes the number of target instances), the insecurity against any adversary $\mathcal{A}$ restricted to a running time ξ is defined as the supremum of its capabilities. Depending on the nature of Prop , this capability is measured either as the maximum success probability in search-based experiments or the maximum advantage in distinguishing experiments. Formally, we express this as:

$$\text{InSec}^{\text{Prop}}(F; \xi, p) = \max_{\mathcal{A}} \left\{ \text{Succ}_{F,p}^{\text{Prop}}(\mathcal{A}), \text{Adv}_{F,p}^{\text{Prop}}(\mathcal{A}) \right\}.$$

We say that an adversary has *tweak advice* if, before the attack phase, it commits to the full set of p target tweaks it intends to attack. This captures a non-adaptive multi-target setting, where all targets are fixed in advance and cannot be chosen adaptively based on intermediate outcomes.

2.2 Multi-Target Preimage Resistance and Its Open Variant

We recall two preimage-resistance notions for tweakable hash functions: the single-function multi-target variant (SM-PRE) and its open variant (SM-openPRE). These notions capture multi-target preimage resistance under a fixed public parameter and distinct tweaks. In SM-PRE, an adversary creates multiple preimage challenges under a fixed public parameter, each associated with a distinct tweak. The single-function setting captures that all challenges share the same parameter, while the multi-target setting allows the adversary to attack several instances simultaneously. The requirement of distinct tweaks enforces domain separation across targets.

SM-openPRE extends this model by granting access to an opening oracle that reveals the hidden preimages for selected targets. This abstraction is particularly suited to settings where most targets become exposed during execution, leaving only a subset to be attacked. In SPHINCS+, this captures the behavior of the FORS component, where only a limited number of targets remain hidden.

Throughout this subsection, let $\mathbf{Th} : \mathcal{P} \times \mathcal{T} \times \mathcal{M} \rightarrow \mathcal{N}$ denote a tweakable hash function. Both notions are parameterized by the number of targets $p \leq |\mathcal{T}|$. We consider a two-stage adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ operating against p targets, where $p \leq |\mathcal{T}|$. In these games, the first stage $\mathcal{A}_1$ sets up the challenge by making exactly p classical queries to a randomized target oracle. For each queried tweak T_i , the oracle internally samples a secret $x_i \xleftarrow{\$} \mathcal{M}$ and returns the evaluation $Y_i = \mathbf{Th}(P, T_i, x_i)$. The public parameter $P \in \mathcal{P}$ remains hidden during this initial phase. Let $Q = \{T_1, \dots, T_p\}$ denote the set of queried tweaks. To enforce rigid domain separation across challenges, we require the predicate $\text{DIST}(Q)$ to evaluate to true, which guarantees that all T_i are mutually distinct (i.e., $T_i \neq T_k$ for all $i \neq k$).

Definition 2 (SM-PRE, [3]). *Upon completing the setup phase, the second stage $\mathcal{A}_2$ receives the internal state from $\mathcal{A}_1$ along with the now-revealed public parameter P . The adversary succeeds if it can output a valid preimage for any of the previously established targets. Formally, the success probability of $\mathcal{A}$ against the SM – PRE property is defined as:*

$$\text{Succ}_{\mathbf{Th},p}^{\text{SM-PRE}}(\mathcal{A}) = \Pr \left[\begin{array}{l} P \xleftarrow{\$} \mathcal{P} \\ (Q, S) \leftarrow \mathcal{A}_1^{\mathbf{Th}(P, \cdot, x_i)}() \\ (j, M) \leftarrow \mathcal{A}_2(Q, S, P) \end{array} : \begin{array}{l} \mathbf{Th}(P, T_j, M) = \mathbf{Th}(P, T_j, x_j) \\ \wedge \text{DIST}(Q) \end{array} \right].$$

Definition 3 (SM-openPRE, [6, 10]). *The SM – openPRE property extends the standard experiment by granting $\mathcal{A}_2$ additional capabilities: access to an opening oracle, denoted as $\text{Open}(\cdot)$. When queried with an index $i \in [1, p]$, this oracle explicitly reveals the corresponding secret x_i . Let Q_{Open} be the set of indices queried to this oracle. The adversary wins if it can forge a preimage for any target that was **not** trivially opened. The success probability is formulated as:*

$$\text{Succ}_{\mathbf{Th},p}^{\text{SM-openPRE}}(\mathcal{A}) = \Pr \left[\begin{array}{l} P \xleftarrow{\$} \mathcal{P} \\ (Q, S) \leftarrow \mathcal{A}_1^{\mathbf{Th}(P, \cdot, x_i)}() \\ (j, M) \leftarrow \mathcal{A}_2^{\text{Open}(\cdot)}(Q, S, P) \end{array} : \begin{array}{l} \mathbf{Th}(P, T_j, M) = \mathbf{Th}(P, T_j, x_j) \\ \wedge \text{DIST}(Q) \\ \wedge j \notin Q_{\text{Open}} \end{array} \right].$$

According to the security analysis of SPHINCS+ [8, 9], the insecurity of both the SM – PRE and SM – openPRE properties can be bounded by reducing them to two other multi-target properties of the underlying tweakable hash function: SM-TCR and SM-DSPR. Formally, prior work [2, 3, 6] establishes the following relationship for an adversary $\mathcal{A}$:

$$\text{Succ}_{\mathbf{Th},p}^{\text{SM-(open)PRE}}(\mathcal{A}) \leq 3 \cdot \text{Succ}_{\mathbf{Th},p}^{\text{SM-TCR}}(\mathcal{B}) + \text{Adv}_{\mathbf{Th},p}^{\text{SM-DSPR}}(\mathcal{C}), \quad (1)$$

where $\mathcal{B}$ and $\mathcal{C}$ are efficient adversaries constructed via polynomial-time reductions.

The conjectural tight bound: The main limitation of Equation 1 lies in bounding the SM-DSPR advantage. In the QROM, the best proven bound scales as $O(p \cdot q^2/2^n)$, i.e., linearly in the number of targets p . Existing SPHINCS+

analyses conjecture that this dependence can be removed, leading to a tighter estimate of order $O(q^2/2^n)$. Consequently, the tight security parameters adopted in SLH-DSA implicitly rely on this conjecture, and the resulting SM-PRE and SM-openPRE bounds inherit this assumption.

The provable loose bound: Without this conjecture, one must instead reduce SM-openPRE to single-target preimage resistance via a guessing argument, obtaining

$$\text{Succ}_{\text{Th},p}^{\text{SM-openPRE}}(\mathcal{A}) \leq p \cdot \text{Succ}_{\text{Th},1}^{\text{PRE}}(\mathcal{A}'). \quad (2)$$

This bound is fully provable but incurs a linear loss in p . In SPHINCS+, where $p = 2^a \cdot k$, this loss becomes significant in concrete parameter settings.

This gap between conjectural tightness and provable looseness motivates our refined, conjecture-free analysis.

When **Th** is modeled as a quantum random oracle, the best-known concrete estimates are

$$\text{Adv}_{\text{Th},p,q}^{\text{SM-TCR}}(\mathcal{A}) \leq 8(2q+1)^2/2^n + 16q^2/|\mathcal{P}|,$$

and

$$\text{Adv}_{\text{Th},p,q}^{\text{SM-DSPR}}(\mathcal{A}) \leq 32pq^2/2^n + 16q^2/|\mathcal{P}|.$$

In the SPHINCS+ setting, $\mathcal{P} = \{0,1\}^n$, so the terms $16q^2/|\mathcal{P}|$ are absorbed into $O(q^2/2^n)$. Hence, the standard analysis yields a conjectural bound of order $\Theta((q+1)^2/2^n)$ for SM-PRE and SM-openPRE. However, this estimate relies on the conjectured improvement of SM-DSPR to a bound independent of the number of targets p . Without this conjecture, the provable reduction for multi-target security becomes looser and incurs an explicit factor proportional to p .

In recent works by Kudinov et al. [13, 14], a conjecture-free bound for SM – PRE was established in QROM. Kudinov et al. introduced the auxiliary notion of single-function multi-target one-wayness (SM – OW) for distinct tweaks. By reducing SM – PRE to SM – OW and SM – UD, they derived a provable QROM bound of the form:

$$\text{Adv}_{\text{Th},\mathcal{M}',p}^{\text{SM-PRE}}(\mathcal{A}) \leq \frac{8(q+1)^2}{|\mathcal{H}|} + \frac{12(q+1)}{\sqrt{|\mathcal{M}'|}},$$

2.3 Average-Case Search Problem

To establish our security bounds in the subsequent sections, we rely on the Average-Case Search (Avg-Search) problem and its known quantum query complexity [1].

Definition 4 (Avg-Search Problem, [1]). *Let D_λ be a distribution over the family of boolean functions $f : \{0,1\}^m \rightarrow \{0,1\}$ such that for any given input x , $f(x) = 1$ with probability $\lambda \in [0,1]$, and $f(x) = 0$ with probability $1 - \lambda$. The Avg-Search $_\lambda$ problem tasks a quantum adversary $\mathcal{A}$, given quantum oracle access to a sampled function $f \leftarrow D_\lambda$, with finding an input x such that $f(x) = 1$. The success probability of any q -query quantum algorithm $\mathcal{A}$ is defined as:*

$$\text{Succ}^{\text{Avg-Search}_\lambda}(\mathcal{A}) = \Pr_{f \leftarrow D_\lambda} \left[f(x) = 1 : x \leftarrow \mathcal{A}^{|f\rangle}(\cdot) \right]$$

The quantum query complexity for this problem is strictly bounded as follows:

Lemma 5 ([1]). *For any quantum algorithm $\mathcal{A}$ making at most q quantum queries to the oracle f , the success probability against the Avg-Search $_\lambda$ problem is bounded by:*

$$\text{Succ}^{\text{Avg-Search}_\lambda}(\mathcal{A}) \leq 8\lambda(q+1)^2$$

2.4 Architecture of the SPHINCS+ (SLH-DSA) Framework

SPHINCS+, recently standardized by NIST as SLH-DSA (FIPS 205), is a stateless hash-based signature scheme. Its security conservatively relies solely on the security properties of the underlying cryptographic hash functions. To provide domain separation and mitigate multi-target attacks, SPHINCS+ uses a family of tweakable hash functions, denoted by **Th**, which combine a public seed with an address (ADRS) that uniquely identifies each hash invocation.

The architecture of SPHINCS+ is constructed from three hierarchical cryptographic components: the Winternitz One-Time Signature (WOTS+), the Forest of Random Subsets (FORS), and a structured Hypertree of Merkle trees (see Figure 1).

Winternitz One-Time Signature: WOTS+ is the one-time signature scheme used throughout SPHINCS+ to authenticate intermediate values, including FORS public keys and subtree roots in the hypertree. Its security is based on the

preimage and collision resistance of the underlying tweakable hash functions. As a one-time scheme, each WOTS+ key pair must be used to sign at most one message.

Forest of Random Subsets: FORS is a few-time signature (FTS) scheme explicitly designed to sign the randomized message digest. A FORS instance consists of k independent binary hash trees, each of height a . The public key of FORS is the hash of the k root nodes of these subtrees. To sign a message, the message digest is parsed into k indices, each determining a specific leaf in one of the k subtrees. The signature consists of the preimages (secret keys) corresponding to these k leaves, along with their respective authentication paths to the roots.

Importantly, a FORS signature exposes exactly k target preimages out of the total $p = k \cdot 2^a$ possible leaves. This structural property is central to our analysis and motivates the refined SM-openPRE bound developed in Section 3.

The Hypertree Structure: To support a large number of signatures without maintaining state, SPHINCS+ organizes its Merkle trees into a hypertree of total height h , divided into d layers. Each layer consists of multiple eXtended Merkle Signature Scheme (XMSS) trees of height h/d . The Hypertree establishes a cryptographic chain of trust from the FORS public key up to the global public key of the SPHINCS+ scheme: the bottom layer authenticates the FORS public key, each higher layer authenticates the roots of the layer below it, and the root of the top layer serves as the public key of SPHINCS+. More precisely,

- The bottom-most layer (Layer 0) contains XMSS trees whose leaves are WOTS+ key pairs used to sign the public keys (roots) of the FORS instances.
- For any intermediate layer, the leaves of its XMSS trees are WOTS+ key pairs used to sign the root nodes of the XMSS trees in the layer directly below it.
- The root node of the single XMSS tree at the top-most layer (Layer $d - 1$) serves as the overarching public key of the SPHINCS+ scheme.

At a high level, SPHINCS+ first maps the message into a compact digest and a small set of indices using a randomized hashing step. The digest is then signed using FORS, which is a few-time signature scheme designed to authenticate short message digests efficiently. The resulting FORS public key is then signed using WOTS+ within the bottom layer of a Merkle tree. Finally, the corresponding Merkle authentication path is provided so that the verifier can recover and authenticate the leaf public key. This process is repeated across multiple layers of a hypertree, where each subtree root is authenticated by a WOTS+ signature in the next higher layer. The topmost Merkle root serves as the public key of the entire scheme.

Signing and verification: To sign a message M , SPHINCS+ first computes a randomized message digest and derives the corresponding indices for the FORS instance and the hypertree path. The signing algorithm then proceeds bottom-up: it signs the digest using FORS, authenticates the resulting FORS public key using WOTS+ at the bottom layer, and continues signing subtree roots layer by layer up to the top level. The final signature consists of the randomness, the FORS signature, and the sequence of WOTS+ signatures and authentication paths across the d layers. Verification reconstructs this chain of authentication and accepts if and only if the computed top-level root matches the public key.

More concretely, the hypertree structure allows SPHINCS+ to authenticate a very large number of one-time or few-time keys without storing or updating any signing state. Each layer of the hypertree contains many small Merkle subtrees, and each subtree root is signed by a WOTS+ instance at the next layer. This recursive construction yields a compact and stateless mechanism for producing many signatures from a single master key. The security of the scheme relies on the collision resistance, preimage resistance, and related properties of the underlying tweakable hash functions, together with the pseudorandomness of the seed-derived secret values.

2.5 Existing Security Bounds of the SPHINCS+

The security of the SPHINCS+ framework is established through a sequence of cryptographic reductions. In particular, SPHINCS+ has been formally proven to achieve Existential Unforgeability under Chosen Message Attacks (EUF-CMA) with a tight reduction to the underlying properties of its THF, $\mathbf{Th}$ and pseudorandom primitives [2, 3]. Specifically, the security of SPHINCS+ relies on the hardness of multi-target problems, bounding the overall insecurity by the sum of advantages against PRF, ITSr, and the single-function multi-target of $\mathbf{Th}$, namely SM-TCR, SM-PRE, and SM-DSPR, as follows.

Lemma 6 (Theorem 3, [2]). *Let n, w, h, d, m, t, k be the standard SPHINCS+ parameters, where n is the security parameter, w is the Winternitz parameter, h is the hypertree height, d is number of hypertree layers, m is the message length, t is number of leaves in each FORS tree, and k is the number of trees in a FORS signature. Let l denote the number of chains in the WOTS+ instances. Then SPHINCS+ is EUF-CMA secure if the following properties hold:*

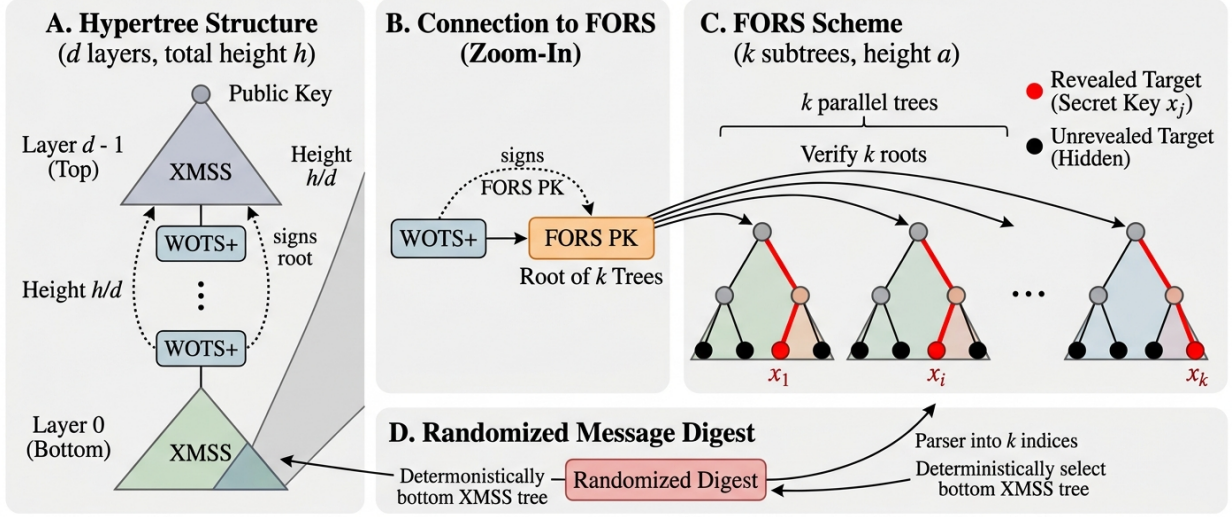


Figure 1: Architecture of SPHINCS+. A: The overall d -layer Hypertree establishing a cryptographic chain of trust from the FORS public key up to the global public key. B: Zoom-in view showing how Layer 0 WOTS+ strictly signs the FORS public key. C: Structured view of FORS scheme composed of k independent trees of height a .

- **Th** (including **F** and **H**) provides single-function multi-target target collision resistance (SM-TCR) with distinct tweaks.
- **F** satisfies single-function multi-target decisional second preimage resistance (SM-DSPR) with distinct tweaks.
- **PRF** and **PRF_{msg}** are post-quantum pseudorandom Function (PRF) families.
- **H_{msg}** satisfies post-quantum interleaved target subset resistance (ITSR).

Moreover, for any adversary $\mathcal{A}$ making at most q_s signing queries, the EUF-CMA insecurity is bounded by

$$\begin{aligned}
 \text{InSec}^{\text{EUF-CMA}}(\text{SPHINCS+}; \xi, q_s) \leq & \\
 & \text{InSec}^{\text{PRF}}(\text{PRF}, \xi, q_1) + \text{InSec}^{\text{PRF}}(\text{PRF}_{\text{msg}}, \xi, q_s) + \\
 & \text{InSec}^{\text{ITSR}}(\text{H}_{\text{msg}}, \xi, q_s) + w \cdot \text{InSec}^{\text{SM-UD}}(\mathbf{F} \in \mathbf{Th}; \xi, q_2) + \\
 & \text{InSec}^{\text{SM-TCR}}(\mathbf{F} \in \mathbf{Th}; \xi, q_3 + q_7) + \text{InSec}^{\text{SM-PRE}}(\mathbf{F} \in \mathbf{Th}; \xi, q_2) + \\
 & \text{InSec}^{\text{SM-TCR}}(\mathbf{H} \in \mathbf{Th}; \xi, q_4) + \text{InSec}^{\text{SM-TCR}}(\mathbf{T}_k \in \mathbf{Th}; \xi, q_5) + \\
 & \text{InSec}^{\text{SM-TCR}}(\mathbf{T}_l \in \mathbf{Th}; \xi, q_6) + \\
 & 3 \cdot \text{InSec}^{\text{SM-TCR}}(\mathbf{F} \in \mathbf{Th}; \xi, q_8) + \text{InSec}^{\text{SM-DSPR}}(\mathbf{F} \in \mathbf{Th}; \xi, q_8),
 \end{aligned}$$

where the number of invoked instances are bounded by: $q_1 < 2^{h+1}(kt + l)$, $q_2 < 2^{h+1} \cdot l$, $q_3 < 2^{h+1} \cdot l \cdot w$, $q_4 < 2^{h+1}k \cdot 2t$, $q_5 < 2^h$, $q_6 < 2^{h+1}$, $q_7 < 2^{h+1}kt$, and $q_8 < 2^h$.

To evaluate this bound concretely, the standard proof proceeds by a game-hopping argument, summarized in Table 3. Starting from the EUF-CMA experiment, the proof successively replaces pseudorandom functions by random functions, isolates the ITSR condition, rules out collisions through SM-TCR reductions, and finally reduces the remaining step to a multi-target preimage-type property.

By aggregating the probability bounds from the game-hopping argument, the comprehensive overarching bound can be summarized as follows:

Game	Modifications from Previous Game	Cryptographic Reduction
GAME_0	The standard EUF-CMA security experiment for SPHINCS+	–
GAME_1	The function PRF is replaced by a random oracle	Reduction to the PRF property of PRF
GAME_2	The function PRF_{msg} is replaced by a random oracle	Reduction to the PRF property of PRF_{msg}
GAME_3	The experiment aborts if the ITSR condition regarding FORS key reuse is violated	Reduction to the ITSR property of $\mathbf{H}_{\text{msg}}$
$\text{GAME}_{3.1}$	Excludes second preimages in the binary hash trees using $\mathbf{H}$	Reduction to SM-TCR for $\mathbf{H}$ (Hyper-tree)
$\text{GAME}_{3.2}$	Excludes second preimages in the compression of FORS root nodes using $\mathbf{T}_k$	Reduction to SM-TCR for $\mathbf{T}_k$ (FORS trees)
$\text{GAME}_{3.3}$	Excludes second preimages in the WOTS+ public keys using $\mathbf{T}_l$	Reduction to SM-TCR for $\mathbf{T}_l$ (FORS PK compression)
$\text{GAME}_{3.4}$	Excludes second preimages in the FORS signature values using $\mathbf{F}$	Reduction to SM-TCR for $\mathbf{F}$ (FORS leaves)
GAME_4	The adversary $\mathcal{A}$ succeeds only if it outputs a preimage for the compression function $\mathbf{F}$ corresponding to a target previously revealed during signature queries (i.e., a WOTS+ signature forgery)	Reduction to the concrete security bound of WOTS+
Bounding GAME_4 (Standard Approach): Evaluating the SM-openPRE property.		Reduces to $3 \cdot \text{SM-TCR} + \text{SM-DSPR}$ for $\mathbf{F}$

Table 3: Summary of the game-hopping sequence in SPHINCS+ security proof

$$\begin{aligned}
\text{InSec}_{\text{SPHINCS}^+}^{\text{EUF-CMA}}(\mathcal{A}) \leq & \underbrace{\text{InSec}^{\text{PRF}} + \text{InSec}^{\text{PRF}}}_{\text{GAME}_1, \text{GAME}_2: \text{Key/Message pseudorandomness}} + \underbrace{\text{InSec}^{\text{ITSR}}}_{\text{GAME}_3: \text{Subset resistance}} \\
& + \underbrace{\sum \text{InSec}^{\text{SM-TCR}}(\mathbf{H}, \mathbf{T}_k, \mathbf{T}_l, \mathbf{F})}_{\text{GAME}_{3.1} - \text{GAME}_{3.4}: \text{Collisions (Hypertree/Roots/PK/Leaves)}} \\
& + \underbrace{\text{InSec}^{\text{WOTS+ factor}}}_{\text{GAME}_4: \text{Same leaf / preimage}}
\end{aligned}$$

Following the NIST evaluation [2, 8, 9], the practical classical and quantum security (tight) bounds can be summarized as:

$$\text{InSec}^{\text{EUF-CMA}}(\text{SPHINCS}^+; q_h) = O\left(\frac{q_h}{2^{8n}} + q_h X\right);$$

and

$$\text{InSec}^{\text{PQ-EUF-CMA}}(\text{SPHINCS}^+; q_h) = O\left(\frac{q_h^2}{2^{8n}} + q_h^2 X\right);$$

where $X = \sum_{\gamma} (1 - (1 - 1/t)^{\gamma})^k \binom{q_s}{\gamma} (1 - 1/2^h)^{q_s - \gamma} \frac{1}{2^{h\gamma}}$ and q_h denotes the number of offline hash queries. In practice, one obtains a concrete bit-security estimate by solving $\text{InSec} \approx 1$ for the maximal feasible q_h and converting q_h to bits of work factor.

The conjectural bottleneck: The main difficulty in the standard evaluation arises in the Game 4 step, where one needs to bound the success probability of forging a preimage for the compression function $\mathbf{F}$. The tight estimate in the literature relies on the conjectural SM-DSPR property. While this yields a numerically sharp bound, SM-DSPR is not known to follow from standard assumptions. If one avoids this conjectural step and seeks a fully provable reduction, the analysis must fall back on SM-PRE. The resulting reduction is typically much looser and incurs a loss factor proportional to the total number of theoretical targets, namely $p = 2^a \cdot k$ for the FORS forest. This gap is the main motivation for our analysis. In the next section, we show that by working through SM-openPRE, we can bypass the conjectural SM-DSPR assumption while paying only for the targets that are actually exposed in practice.

3 A New Perspective on the Security Bounds of SPHINCS+

This section presents our alternative analysis of the SPHINCS+ security bounds. We first establish a provable bound for SM-PRE and then derive an optimized reduction for SM-openPRE, which together provide a conjecture-free evaluation of the SPHINCS+ security parameters.

3.1 An Approach to Evaluating SM-PRE

In the same spirit as Kudinov et al. [13, 14], who avoid conjectural assumptions, our approach follows a different reduction strategy by relating SM-PRE to Avg-Search problem [1], yielding the provable bound $\text{Succ}_{\text{Th},p}^{\text{SM-PRE}}(\mathcal{A}) \leq 32(q + 1/2)^2/2^n$.

We now establish a conjecture-free security bound for the SM-PRE. To achieve this, we construct an adversary $\mathcal{B}$ that solves the Average-Case Search (Avg-Search) problem by leveraging an adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ that attacks SM-PRE. Building upon the proof techniques for SM-TCR introduced in [2], we evaluate the SM-PRE bound under the realistic scenario where the initial phase adversary, $\mathcal{A}_1$, is restricted to classical oracle queries.

Lemma 7. *Let Th be a THF. Any quantum adversary $\mathcal{A}$ that breaks the SM-PRE for p targets by making q quantum queries to Th can be used to construct a quantum adversary $\mathcal{B}$ making at most $2q$ queries that solves the Avg-Search $_\lambda$ problem (with $\lambda = 1/2^n$) with a success probability:*

$$\text{Succ}^{\text{Avg-Search}_\lambda}(\mathcal{B}) \geq \text{Succ}_{\text{Th},p}^{\text{SM-PRE}}(\mathcal{A})$$

Proof. We construct the algorithm $\mathcal{B}$ to solve the Avg-Search problem by simulating the SM-PRE environment for $\mathcal{A}$ as follows:

Simulation of $\mathcal{A}_1$: Initially, $\mathcal{B}$ simulates the interaction with $\mathcal{A}_1$. The adversary $\mathcal{A}_1$ issues p classical queries to the challenge oracle with distinct tweaks $\{T_1, \dots, T_p\}$. For each query T_i , $\mathcal{B}$ must return a target value y_i . $\mathcal{B}$ proceeds by:

1. Uniform random sampling an output value $y_i \xleftarrow{\$} \{0, 1\}^n$. (In the SM-PRE experiment, this corresponds to $\text{Th}(P, T_i, x_i)$. Since x_i is hidden from $\mathcal{A}_1$ and Th outputs are assumed to be uniformly distributed, $\mathcal{B}$ perfectly simulates this without needing x_i).
2. Returning y_i to $\mathcal{A}_1$. $\mathcal{B}$ then stores the target list $Q = \{(T_i, y_i)\}_{i=1}^p$ and associates each target with the index i of the challenge function f .

Parameter Setup and Oracle Programming ($\widetilde{\text{Th}}$): After $\mathcal{A}_1$ completes its queries, $\mathcal{B}$ samples a public parameter $P \xleftarrow{\$} \mathcal{P}$ and receives an instance of the Avg-Search challenge, characterized by a function $f \leftarrow D_\lambda : [p] \times \{0, 1\}^\alpha \rightarrow \{0, 1\}$ with $\lambda = 1/2^n$. $\mathcal{B}$ constructs a simulated quantum oracle $\widetilde{\text{Th}}$ for the second phase adversary $\mathcal{A}_2$ as follow.

1. For each $i \in [1, p]$, $\mathcal{B}$ chooses a random function $g_i : \{0, 1\}^\alpha \rightarrow \{0, 1\}^n \setminus \{y_i\}$ and a global random function $h : \mathcal{P} \times \mathcal{T} \times \{0, 1\}^\alpha \rightarrow \{0, 1\}^n$.
2. For any input query $(P^*, T^*, M^*) \in \mathcal{P} \times \mathcal{T} \times \{0, 1\}^\alpha$, the oracle $\widetilde{\text{Th}}$ is programmed as follows:
 - (a) If $T^* = T_i \wedge P^* = P$ for some $i \in [1, p]$:

$$\widetilde{\text{Th}}(P^*, T^*, M^*) = \begin{cases} y_i & \text{if } f(i, M^*) = 1 \\ g_i(M^*) & \text{if } f(i, M^*) = 0 \end{cases}$$

- (b) If $T^* \notin \{T_1, \dots, T_p\} \vee P^* \neq P$:

$$\widetilde{\text{Th}}(P^*, T^*, M^*) = h(P^*, T^*, M^*)$$

Solution Extraction and Correctness Analysis: $\mathcal{B}$ runs $\mathcal{A}_2(Q, S, P)$ with quantum access to the programmed oracle $\widetilde{\text{Th}}$. Eventually, $\mathcal{A}_2$ outputs a pair (j, M^*) , claiming that M^* is a preimage for the j -th target T_j . $\mathcal{B}$ immediately outputs (j, M^*) as its solution to the Avg-Search challenge.

To analyze correctness, note that $\widetilde{\text{Th}}$ perfectly simulates the distribution of a real SM-PRE experiment because the probability that $f(j, M^*) = 1$ is exactly $1/2^n$, seamlessly matching the probability that a random input M^*

yields the specific hash digest y_i . According to the SM-PRE winning condition, $\mathcal{A}$ succeeds if it finds an M^* such that $\widetilde{\text{Th}}(P, T_j, M^*) = y_j$. Based on our oracle programming in Case (a), this condition is satisfied if and only if $f(j, M^*) = 1$. Therefore, whenever $\mathcal{A}$ outputs a valid preimage, (j, M^*) is guaranteed to be a solution to the Avg-Search problem. Consequently, $\text{Succ}^{\text{Avg-Search}_\lambda}(\mathcal{B}) \geq \text{Succ}_{\text{Th}, p}^{\text{SM-PRE}}(\mathcal{A})$. $\square$

Security Bound for SM-PRE: The quantum query complexity bound for the Avg-Search $_\lambda$ problem is well-established in [1]. For an adversary making at most $2q$ quantum queries, the success probability is strictly bounded by:

$$\text{Succ}^{\text{Avg-Search}_\lambda}(\mathcal{B}) \leq 8\lambda(2q+1)^2 = 32(q+1/2)^2/2^n$$

By combining this with Lemma 7, we establish the provable security bound for SM-PRE.

3.2 Optimized Bound for SM-openPRE via Tight Reduction

In the SPHINCS+ framework, the SM-openPRE property is centrally utilized to evaluate the security of the FORS signature component. The total number of targets corresponding to the SM-openPRE instance is $p = 2^h \cdot k \cdot 2^a$ (where h is the hypertree height, k is the number of FORS subtrees, and a is the height of each subtree). However, in practice, at most $p' = q_s \cdot k$ targets are actually exposed through queries during signature generation, where q_s is the maximum number of signatures (typically a divisor of 2^{h+a}).

Consequently, we evaluate an SM-openPRE instance with p targets, but restrict the adversary to at most p' queries to the $\text{Open}(\cdot)$ oracle. To formalize this, let $\text{Succ}_{\text{Th}, p, p'}^{\text{SM-openPRE}}(\mathcal{A})$ denote the success probability of an adversary $\mathcal{A}$ operating under these specific constraints.

Proposition 8. *Let $p = c \cdot p'$ (where c and p' are positive integers, and $c \ll p$). Let $\mathcal{A}$ be an adversary against the SM-openPRE of a tweakable hash function Th with p targets, making at most q standard queries and at most p' queries to the $\text{Open}(\cdot)$ oracle. Then, there exists an adversary $\mathcal{B}$ against the SM-PRE with c targets, making at most $q + p - c$ queries, such that:*

$$\text{Succ}_{\text{Th}, p, p'}^{\text{SM-openPRE}}(\mathcal{A}) \leq e \cdot p' \cdot \text{Succ}_{\text{Th}, c}^{\text{SM-PRE}}(\mathcal{B})$$

where $e \approx 2.718$ is Euler's number.

Proof. We construct a reduction algorithm $\mathcal{B} = \mathcal{M}^{\mathcal{A}}$ that solves the SM-PRE problem (with c targets) by simulating the SM-openPRE environment for the adversary $\mathcal{A}$ (that making at most p' queries to the opening oracle $\text{Open}(\cdot)$) as follows:

Reduction $\mathcal{M}^{\mathcal{A}}$:

1. $\mathcal{M}^{\mathcal{A}}$ begins the SM-PRE game (with c targets) by receiving p tweak advices $\{T_i\}_{i=1}^p$ from $\mathcal{A}$.
2. $\mathcal{M}^{\mathcal{A}}$ randomly selects c distinct indices from $\{1, \dots, p\}$, denoted as $i_1^*, \dots, i_c^*$. It then sends queries $\{T_{i_j^*}\}_{j=1}^c$ to its $\text{Th}(P, \cdot, x_{i_j^*})$ oracle to receive the values $y_{i_j^*}$, where $x_{i_j^*}$ is chosen randomly according to the j th query of $\mathcal{M}^{\mathcal{A}}$.
3. For the remaining tweak values, $\mathcal{M}^{\mathcal{A}}$ first enters Stage 2) to receive the public value P . Then, for each index $i \notin \{i_1^*, \dots, i_c^*\}$, $\mathcal{M}^{\mathcal{A}}$ randomly chooses x_i and queries (P, T_i, x_i) to $\text{Th}(\cdot, \cdot, \cdot)$ to obtain the value y_i .
4. $\mathcal{M}^{\mathcal{A}}$ returns the values y_i obtained above for each tweak T_i to $\mathcal{A}$, for all $i \in \{1, \dots, p\}$.
5. $\mathcal{M}^{\mathcal{A}}$ initiates Stage 2) of $\mathcal{A}$ by providing the public value P . For every query (P', T', M') from $\mathcal{A}$ to $\text{Th}(\cdot, \cdot, \cdot)$, $\mathcal{M}^{\mathcal{A}}$ acts as a forwarder, passing the query to the $\text{Th}(\cdot, \cdot, \cdot)$ oracle and returning the response to $\mathcal{A}$.
6. If $\mathcal{A}$ requests a query to the $\text{Open}(\cdot)$ oracle for a tweak index $i \notin \{i_1^*, \dots, i_c^*\}$, $\mathcal{M}^{\mathcal{A}}$ returns the value x_i ; otherwise, $\mathcal{M}^{\mathcal{A}}$ aborts the SM-openPRE experiment.
7. If $\mathcal{A}$ outputs (i^*, x^*) , then $\mathcal{M}^{\mathcal{A}}$ outputs (i^*, x^*) for the SM-PRE experiment.

Simulation success probability analysis: Let Sim denotes the event of a perfect simulation where $\mathcal{M}^{\mathcal{A}}$ successfully avoids any abort conditions during $\mathcal{A}$'s queries. We see that $\mathcal{M}^{\mathcal{A}}$ perfectly simulates the SM-openPRE experiment if the p ($\leq p'$) queries to the $\text{Open}(\cdot)$ oracle made by $\mathcal{A}$ fall on indices outside the set $\{i_1^*, \dots, i_c^*\}$. Since the set $\{i_1^*, \dots, i_c^*\}$ is chosen randomly, there are $\binom{p-c}{p}$ ways to choose p indices from the set $\{1, \dots, p\} \setminus \{i_1^*, \dots, i_c^*\}$ to ensure a perfect

simulation. Meanwhile, there are $\binom{p}{p}$ possible choices that could be made by $\mathcal{A}$ for the p queries to the $\text{Open}(\cdot)$ oracle. Thus, the probability that $\mathcal{M}^{\mathcal{A}}$ successfully simulates the SM-openPRE experiment for $\mathcal{A}$ is:

$$\begin{aligned} \Pr[\text{Sim}] &= \frac{\binom{p-c}{p}}{\binom{p}{p}} = \frac{(p-c-p+1) \dots (p-p)}{(p-c+1) \dots p} \\ &\geq \frac{(p-c-p'+1) \dots (p-p')}{(p-c+1) \dots p} \\ &= \left(1 - \frac{p'}{p-c+1}\right) \dots \left(p - \frac{p'}{p}\right) \end{aligned}$$

Since $c \ll p$ and $p = c \cdot p'$, we can approximate the terms as:

$$\Pr[\text{Sim}] \geq \left(1 - \frac{p'}{p-c+1}\right)^c \approx \left(1 - \frac{p'}{p}\right)^c = \left(1 - \frac{1}{c}\right)^c \approx \frac{1}{e}$$

Winning condition: According to the simulation, $\mathcal{A}$ outputs a result (i^*, x^*) only if i^* is not an index for which $\mathcal{A}$ previously requested an $\text{Open}(\cdot)$ query. In other words, in this scenario, $\mathcal{M}^{\mathcal{A}}$ accurately simulates the SM-openPRE experiment. Furthermore, if (i^*, x^*) is a successful output for $\mathcal{A}$, it is also a successful output for $\mathcal{M}^{\mathcal{A}}$ provided that $i^* \in \{i_1^*, \dots, i_c^*\}$. Let I^* denotes the set of $\{i_1^*, \dots, i_c^*\}$. Since I^* is chosen uniformly at random and is independent of $\mathcal{A}$'s view (conditioned on no aborts), the probability that $i^* \in I^*$ is at least $\frac{c}{p} = \frac{1}{p'}$. Consequently, the overall success probability of $\mathcal{B} = \mathcal{M}^{\mathcal{A}}$ is bounded by:

$$\begin{aligned} \text{Succ}_{\text{Th},c}^{\text{SM-openPRE}}(\mathcal{B}) &\geq \Pr[\text{Sim}] \cdot \Pr[i^* \in I^*] \cdot \text{Succ}_{\text{Th},p,p'}^{\text{SM-openPRE}}(\mathcal{A}) \\ &\geq \frac{1}{e} \cdot \frac{1}{p'} \cdot \text{Succ}_{\text{Th},p,p'}^{\text{SM-openPRE}}(\mathcal{A}) \end{aligned}$$

Rearranging the terms yields the final bound:

$$\text{Succ}_{\text{Th},p,p'}^{\text{SM-openPRE}}(\mathcal{A}) \leq e \cdot p' \cdot \text{Succ}_{\text{Th},c}^{\text{SM-openPRE}}(\mathcal{B})$$

□

Remark 9. For SPHINCS+ parameters one typically has $p' \ll p$ and $p' \mid p$; Proposition 8 therefore replaces the earlier p -factor loss with the much smaller p' -factor (up to the constant e), yielding a substantially tighter and practically meaningful bound for the FORS-related SM-openPRE step.

3.3 Impact on the SPHINCS+ Security Bound

Using the SM-PRE bound (Lemma 7) and the SM-openPRE to SM-PRE reduction (Proposition 8) developed above, we can replace the prior (conjectural) dependence on SM-DSPR by rigorous bounds. This removal of conjectural assumptions is achieved at the cost of a tightness loss factor: the reduction for the FORS/GAME₄ step incurs an extra factor proportional to the number of Open-queries actually made during signing, which for SPHINCS+ equals at most $p' = q_s \cdot k$. In other words, the dominant SM-DSPR term is replaced by SM-openPRE/SM-PRE terms with an extra $q_s \cdot k$ loss factor. Consequently, the insecurity bound is reformulated as follows:

Corollary 10 (Conjecture-Free Security Bound of SPHINCS+). *For the parameters n, w, h, d, m, t, k and l as described in the standard SPHINCS+ specifications, let $\mathcal{A}$ be an adversary making at most q_s signature queries. The EUF-CMA insecurity of the scheme is bounded by:*

$$\begin{aligned} \text{InSec}^{\text{EUF-CMA}}(\text{SPHINCS+}; \xi, q_s) &\leq \text{InSec}^{\text{PRF}}(\text{PRF}, \xi, q_1) + \text{InSec}^{\text{PRF}}(\text{PRF}_{\text{msg}}, \xi, q_s) \\ &\quad + \text{InSec}^{\text{ITSR}}(\mathbf{H}_{\text{msg}}, \xi, q_s) + w \cdot \text{InSec}^{\text{SM-UD}}(\mathbf{F} \in \text{Th}; \xi, q_2) \\ &\quad + \text{InSec}^{\text{SM-TCR}}(\mathbf{F} \in \text{Th}; \xi, q_3 + q_7) + \text{InSec}^{\text{SM-PRE}}(\mathbf{F} \in \text{Th}; \xi, q_2) \\ &\quad + \text{InSec}^{\text{SM-TCR}}(\mathbf{H} \in \text{Th}; \xi, q_4) + \text{InSec}^{\text{SM-TCR}}(\mathbf{T}_k \in \text{Th}; \xi, q_5) \\ &\quad + \text{InSec}^{\text{SM-TCR}}(\mathbf{T}_l \in \text{Th}; \xi, q_6) + 3 \cdot \text{InSec}^{\text{SM-TCR}}(\mathbf{F} \in \text{Th}; \xi, q_8) \\ &\quad + \text{InSec}^{\text{SM-openPRE}}(\mathbf{F} \in \text{Th}; \xi, q_8, q_9), \end{aligned}$$

where the number of invoked instances are bounded by: $q_1 < 2^{h+1}(kt + \ell)$, $q_2 < 2^{h+1}\ell$, $q_3 < 2^{h+1}\ell w$, $q_4 < 2^{h+1}k \cdot 2t$, $q_5 < 2^h$, $q_6 < 2^{h+1}$, $q_7 < 2^{h+1}kt$, and q_8 is as in the underlying SPHINCS+ proof; in our setting, the number of opening queries additionally satisfies $q_9 \leq q_s \cdot k$ (i.e., the exact maximum number of targets actually revealed to the adversary across all q_s signature queries).

Table 4: Security comparison for SLH-DSA parameter sets

	n	h	d	h'	a	k	w	m	q_s	classical			quantum		
										NIST (tight)	NIST (loose)	This work	NIST (tight)	NIST (loose)	This work
SLH-DSA-128s	16	63	7	9	12	14	16	34	2^{64}	127	45	60	63	20	27
SLH-DSA-128f	16	66	22	3	6	33	16	30	2^{64}	127	45	58	63	20	26
SLH-DSA-192s	24	63	7	9	14	17	16	42	2^{64}	191	106	123	95	50	59
SLH-DSA-192f	24	66	22	3	8	33	16	39	2^{64}	191	107	122	95	51	58
SLH-DSA-256s	32	64	8	8	14	22	16	49	2^{64}	255	169	187	127	82	91
SLH-DSA-256f	32	68	17	4	9	35	16	47	2^{64}	254	168	186	127	81	90

Security Bounds for SPHINCS+: We integrate our Proposition 1 and Lemma 7 into the classical and quantum bounds. Let $X = \sum_{\gamma} (1 - (1 - \frac{1}{t})^{\gamma})^k \binom{q_s}{\gamma} (1 - \frac{1}{2^n})^{q_s - \gamma} 2^{-h\gamma}$ be the structural probability of a FORS forgery.

Classical Security Bound Instantiating the Avg-Search and SM-openPRE bounds yields the following classical asymptotic form:

$$\text{InSec}^{\text{EUF-CMA}}(\text{SPHINCS+}; q_h) \leq O\left(\frac{q_h(q_s \cdot k + w + 7)}{2^{8n}} + q_h X\right)$$

Solving $\text{InSec} \approx 1$ for q_h gives the security bits $b = \log_2(q_h)$

Quantum Security Bound For the quantum setting the tightened bound becomes (constants shown for clarity)

$$\begin{aligned} \text{InSec}^{\text{PQ-EUF-CMA}}(\text{SPHINCS+}; q_h) &\leq O(12(w + 2) \frac{q_h}{2^{4n}} \\ &\quad + 32(q_s \cdot k + 8) \frac{(q_h + 1/2)^2}{2^{8n}} + q_h^2 \cdot X) \end{aligned}$$

Setting $\text{InSec}^{\text{PQ-EUF-CMA}} \approx 1$ yields a quadratic equation

$$q_h^2 \cdot \left(X + \frac{32(q_s \cdot k + 8)}{2^{8n}}\right) + q_h \cdot \left(\frac{12(w + 2)}{2^{4n}} + \frac{32(q_s \cdot k + 8)}{2^{8n}}\right)$$

≈ 1 , so the security bits are determined by $b_Q = \log_2(q_h)$.

4 Discussion

Table 4 compares three concrete-security evaluations for six FIPS-205 (SLH-DSA) [9] parameter sets: (i) NIST (tight) — the tight reductional estimate reported by NIST; (ii) NIST (loose) — the conservative, looser proven bound; and (iii) This work — the bounds obtained by replacing the SM-DSPR conjectural with our provable SM-PRE/SM-openPRE reductions (Sections 3.1 and 3.2).

When avoiding conjectural assumptions, evaluations must rely on the NIST “loose” bound, which suffers a severe tightness loss proportional to the total targets p . By modeling the adversary’s exact capabilities via the SM-openPRE property, our approach restricts this penalty strictly to the actually exposed targets $(q_s \cdot k)$. Consequently, our rigorous bound significantly bridges the gap, recovering up to 18 bits of classical security and 9 bits of quantum security (e.g., SLH-DSA-256s) compared to the NIST loose evaluation.

Although our computed levels are numerically lower than the NIST “tight” bound, it is crucial to emphasize that the tight bound relies on an assumption regarding ideal SM-DSPR. In contrast, our method provides a conjecture-free, provable cryptographic guarantee. By completely eliminating heuristic reliance while actively mitigating the worst-case tightness loss, our work establishes an acceptable trade-off for evaluating the security bounds of SPHINCS+ parameters.

Example 11 (*Parameter generation, [12]*). As an illustration, using our conjecture-free bounds, we derive a SPHINCS+ parameter set achieving NIST security level 5, namely

$$(n, h, d, h', a, k, w, m, q_s) = (41, 66, 2, 31, 12, 32, 16, 57, 2^{64})$$

This example shows that our analysis can guide parameter selection while avoiding reliance on the SM-DSPR conjecture.

5 Conclusion

In this paper, we presented a refined, tighter security reductions for the SPHINCS+ scheme. By replacing the heuristic assumptions associated with the SM-DSPR with a reduction leveraging the SM-openPRE, we successfully eliminated the reliance on multi-target resistance. Our approach mitigates the severe tightness penalty inherent in previous worst-case evaluations by confining the loss factor strictly to the maximum number of exposed targets. Evaluations on the standardized FIPS 205 parameter sets show that our proven bounds outperform the NIST loose bounds, reclaiming substantial classical and quantum security bits. This work provides a conjecture-free, provable bounds that reflects the practical security of SPHINCS+, offering a reliable framework for future parameter tuning and standardization efforts.

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References

- [1] Hülsing, A., Rijneveld, J., Song, F. “Mitigating multi-target attacks in hash-based signatures”. In Public-Key Cryptography–PKC, Taiwan, March 6-9, 2016, Proceedings, Part I (pp. 387-416).
- [2] Bernstein, D. J., Hülsing, A., Kölbl, S., Niederhagen, R., Rijneveld, J., Schwabe, P. “The SPHINCS+ signature framework”. In Proceedings of the 2019 ACM SIGSAC conference on computer and communications security (pp. 2129-2146).
- [3] Hülsing, A., Kudinov, M. “Recovering the tight security proof of SPHINCS+”. In International Conference on the Theory and Application of Cryptology and Information Security 2022 Dec 5 (pp. 3-33), Springer.
- [4] Bernstein, D. J., Hülsing, A. “Decisional second-preimage resistance: When does SPR imply PRE?”. In International Conference on the Theory and Application of Cryptology and Information Security 2019, Nov. (pp. 33-62), Springer.
- [5] P. Shor. “Polynomial-time algorithm for prime factorization and discrete logarithm on a quantum computer”, SIAM J. Comput., vol.26, no.5, pp. 1484-1509, 1997..
- [6] Barbosa, Manuel, et al. “A Tight Security Proof for SPHINCS+, Formally Verified.” International Conference on the Theory and Application of Cryptology and Information Security: Springer Nature Singapore, 2024 Dec 10 (pp. 35-67).
- [7] <https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization>.
- [8] Jean-Philippe Aumasson, Daniel J. Bernstein, et al. “SPHINCS+ – Submission to the 3rd round of the NIST postquantum project. v3.1”, 2022
- [9] FIPS 205, Stateless Hash-Based Digital Signature Standard <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.204.pdf>
- [10] Yuan, Quan, Mehdi Tibouchi, and Masayuki Abe. “Quantum-access security of hash-based signature schemes.” Australasian Conference on Information Security and Privacy. Cham: Springer, 2023.
- [11] Xagawa, Keita, and Takashi Yamakawa. “(Tightly) QCCA-secure key-encapsulation mechanism in the quantum random oracle model.” International Conference on Post-Quantum Cryptography. Cham: Springer, 2019.
- [12] <https://github.com/danhvuba/SLH-DSA-parameters.git>
- [13] Drake, Justin, Dmitry Khovratovich, Mikhail Kudinov, and Benedikt Wagner. “Hash-based multi-signatures for post-quantum ethereum.” Cryptology ePrint Archive (2025).
- [14] Kudinov, Mikhail. “Hash Functions in a Post-Quantum World.” Phd Thesis (2025).